Nicola Fontaine

Data Engineer · Data Analyst · Data Scientist · BI, Analytics & Machine Learning · with a B2B sales background

nicolax.fontaine@gmail.com | WhatsApp: +33 6 09 42 40 53 | linkedin.com/in/nicola-fontaine
Portfolio: lunistache.github.io/portfolio | Languages: FR / ES / IT / EN

CORE SKILLS

Languages	Python (pandas, PySpark, scikit-learn), SQL (T-SQL, PostgreSQL), DAX, PowerShell, TypeScript / JavaScript, Node.js
Data Engineering	Design, development and maintenance of robust, efficient data pipelines (ETL / ELT), orchestration & scheduling, error handling & logging, data modelling, distributed processing (Spark)
Data Quality & Integrity	Cleansing, deduplication & normalisation · business rule validation · root-cause investigation of inconsistencies · troubleshooting analytics systems · documentation of data processes and workflows
Data Science & ML	Statistical & exploratory analysis, predictive / analytical model development, feature engineering, model evaluation, financial risk modelling · LLM API integration & prompt engineering
Cloud & Platforms	Azure (Data Factory, Azure SQL Database, Synapse, VM provisioning), Databricks, AWS, Microsoft SQL Server, Microsoft Dataverse, Odoo ERP (V7 / V18)
BI & Analytics	Power BI (data modelling, advanced DAX, report & UX design), Apache Superset (deployment, administration, custom plugins), Tableau, dashboard design
Integration	REST APIs (Mailchimp, Finnhub), SOAP / SoapUI, SDK integration (Avantage), CSV / Excel ingestion & column mapping
Ways of working	DevOps & CI/CD, Git, Agile delivery, client-facing consulting, requirement gathering, user training
Business & Sales	B2B business development, needs discovery, proposals & negotiation, key accounts & public tenders, HubSpot CRM (certified), FR / IT / ES client interface

PROFESSIONAL EXPERIENCE

Data / Power BI Consultant (freelance), Koïno
Oct. 2026 - Present

Engagement for an international oilfield services group, on an Odoo ERP spread across three databases (V7, V18 Community and V18 Enterprise).

- Explored and modelled the Odoo data directly in PostgreSQL: analysed the schema and the standard and custom modules, and handled its specifics (multi-company, multi-currency, translated fields, legacy inherited from earlier versions).
- Designed and built Power BI management reports: employee and equipment certification tracking, purchasing analysis (multi-currency conversion, open orders, receipts, product usage), billing rate and team time allocation.
- Set up data quality checks detecting after-the-fact invoicing, unprocessed receipts, planning over-allocations and inconsistent entries, to give an objective view of process gaps between entities.
- Optimised the data models and DAX measures to stay within the Power BI service's performance limits.
- Scoped the consolidation of the three Odoo databases and the migration of existing reports from V7 to V18.
- Direct point of contact for the client: requirements gathering, regular progress reviews and improvement proposals.

Consultant (V.I.E), SBI (Data | AI | EPM)
Sept. 2025 - Present

Consultant within a Data, AI & EPM team, delivering cloud migration, data pipeline automation, analysis and Power BI reporting projects for large enterprise clients.

Cloud migration & unification: Medifis

- Ran discovery with the client's business and IT teams to map two heterogeneous on-premise data sources, agreed what the unified model had to answer, and proposed the target architecture.
- Designed and provisioned the target Azure environment (servers and Azure SQL Database) to consolidate both sources into a single analytical platform.
- Built the pipelines in Azure Data Factory (Copy Data activities, transformation scripts, Truncate & Reload stored procedures), harmonising the source schemas into layered target tables (staging, then curated tables modelled for BI consumption).
- Wrote and optimised advanced T-SQL (window functions, CTEs, complex joins, stored procedures), tuning the heaviest queries as data volumes grew.
- Implemented full orchestration (triggers, dependencies, scheduling, error handling and logging) so the chain runs reliably and can be safely reprocessed, and documented the source-to-target mapping for business and technical reviewers.

Data cycle automation & quality: Climater

- Replaced a fully manual recurring accounting data cycle: PowerShell scripts using the Avantage SDK extract the data from the client's ERP into an Azure staging environment, where it is transformed at scale with PySpark.
- Designed cleansing and normalisation routines (deduplication, correction of business inconsistencies) and validation rules that caught bad records at the ingestion boundary, safeguarding data integrity.
- Configured the target schemas and relationships in Microsoft Dataverse, automated the ETL for recurring refreshes and deployed it into production, then kept supporting it as the client's rules changed.

Power BI Designer: Frida (Consumer Goods)

- Worked on the client's PostgreSQL database: created views to structure and expose the data consumed by the reports, and cleaned and normalised source data to guarantee its reliability.
- Gathered requirements directly with business stakeholders, then designed the data model, advanced DAX measures, page structure and UX of bespoke Power BI reports over successive iterations.

Engineering practices

- Versioned the work in Azure DevOps with Git (branches, pull requests, code review) and used GitHub Copilot and Claude Code to accelerate script development, refactoring and documentation.

Data Engineer / Data Analyst, Tiepolo

Jan. 2024 - Apr. 2025

Asset management · financial data

- **Database & modelling:** owned the maintenance and evolution of the company's Microsoft SQL Server databases; rebuilt the data models as a documented, layered stack with one agreed definition per metric, ending recurring disputes between teams over whose number was correct.
- **SQL performance:** wrote and optimised advanced T-SQL (window functions, CTEs, complex joins, stored procedures) and automated the refresh of key tables to guarantee integrity and availability.
- **Python ETL pipelines:** built ingestion pipelines pulling financial market data from the Finnhub API and e-mailing campaign data from the Mailchimp API, with incremental loads, validation and error handling, results surfaced in Power BI.
- **Self-service analytics:** introduced Apache Superset in-house on Linux (installation, configuration, administration, custom Python plugins), built reusable datasets so analyses stopped being rebuilt from scratch, and trained the whole team.
- **Reporting:** designed dynamic Power BI dashboards tailored to each portfolio manager's decisions, on shared metric definitions, giving targeted visibility on key performance indicators.
- **Troubleshooting:** investigated figures questioned by portfolio managers, traced discrepancies back to source data or transformation logic, and fixed them in the model rather than patching reports.
- **Internal tooling:** built custom in-house tools for financial analysis and reporting, integrating DevOps and Git practices for reviewable, reproducible work.

Consultant, Adone Conseil

2023

Consulting firm · luxury sector · Hermès Italia and Chanel France

- **Hermès Italia, shoe atelier (near Milan):** on site, analysed existing processes, systems and data flows and found that raw material intake, leather receipt, finished product shipping, discarded materials and unsellable footwear were all logged by hand in separate spreadsheets.
- With my team, designed and built a bespoke Business Intelligence tool from scratch that centralised all these flows, drastically cutting the time spent managing and cross-referencing spreadsheets.
- Led the roll-out end to end: managed the transition, trained the team and kept tuning the tool to evolving business needs.
- **Chanel France:** managed the relationship with the Italian raw-material suppliers of the ready-to-wear, leather goods and footwear lines: order tracking, handling non-conformities and delivery delays, negotiating lead times in Italian and securing supply for the collections.

Data Consultant, ARHIS Competency Center

2022

- Built an internal application (Node.js / Java, SQL, SOAP / SoapUI) to streamline consultants' use of data quality and master data tools (Informatica MDM Hub, IDQ Developer), from requirements gathering through to delivery.
- Conducted in-depth analysis of data requirements and modelled the data to ensure quality, accessibility and a structured, easily understandable representation of the information.
- Applied DevOps practices to ensure smooth collaboration across development, operations and systems maintenance.

Data Analyst, Groupe Osmozium

2021

Assigned to Crédit Agricole Corporate & Investment Bank

- Collected, cleaned and analysed large financial datasets in SQL and Python to surface meaningful trends and patterns.
- Developed statistical and scoring models to assess financial risk on investment projects and support investment decisions.
- Produced detailed reports and visual presentations turning quantitative findings into clear recommendations for internal stakeholders and finance teams.

SELECTED PROJECT

Kalbon, carbon footprint estimator for events 2026, ongoing

Personal project, designed and built end to end

- Multi-tenant SaaS web application that estimates the carbon footprint of an event before it takes place, from around six inputs covering attendee travel, catering, venue energy and logistics.
- Built the calculation engine as a deterministic, dependency-free, unit-tested TypeScript package: it resolves each emission factor through a priority cascade (measured data, then proprietary database, then public ADEME / DEFRA factors) and records the source and version behind every figure.
- Designed the PostgreSQL data model (multi-tenant isolation, versioned factor sets, immutable estimate snapshots) and a CSV / Excel import with column mapping, reusable templates and offline geocoding.
- Integrated an LLM API for natural-language input parsing and result explanation, with prompt engineering, schema validation and guardrails so a malformed model response degrades safely instead of producing a wrong figure. Stack: Next.js 15, TypeScript, PostgreSQL / Drizzle ORM, Tailwind, Recharts · bilingual FR / EN.

BUSINESS DEVELOPMENT & SALES

Co-founder & Head of Sales, Nima Studio 2025 - Present

Website design agency · co-founded with a business partner

- Own 100% of business development: target selection, direct outreach (field, phone, email, LinkedIn, referral) and building the client portfolio from scratch.
- Run discovery meetings to turn the client's objectives into a clear specification and a costed solution; write proposals and quotes, negotiate and close, converting part of each sale into recurring maintenance contracts.
- Single point of contact after signature (change requests, incidents, deadlines) and account growth through upselling (redesign, SEO, hosting).

Technical Sales Representative, ESA - Ecologia Soluzione Ambiente 2024 - 2026

Italian manufacturer of waste management equipment · France market, reporting to the CEO

- Sold the full range (automated collection systems, recycled HDPE and underground containers, solar compactors, water treatment plants) to local authorities and key accounts such as Total and Veolia.
- **Full sales cycle:** project detection, needs qualification, on-site demonstrations, costed offers built with the plant (configuration, price, margin, lead times), negotiation through to signature, and public tender responses.
- Represented ESA at trade shows, followed the portfolio after the sale (commissioning, user training, claims) and acted as the FR / IT / ES bridge with the Italian plant, with pipeline reporting to the CEO.

Technical Sales Representative, Adefia 2021 - 2024

French supplier of industrial storage and environmental solutions · Paris area

- Sold retention containers, safety cabinets, fuel tanks and waste collection equipment (standard and custom-built) to industrial companies and local authorities: prospecting, on-site needs analysis, quotes with the design office, negotiation, closing and after-sales follow-up.

Manager, Entropie Music (alongside studies and employment) Feb. 2015 - Jan. 2026

- Ran a music project for 11 years: built a database of venues and festivals in France and abroad for systematic outreach, sold dates and negotiated fees and performance contracts, organised tours, and managed invoicing, bookkeeping, the website and online store.

EDUCATION, CERTIFICATIONS & LANGUAGES

Master of Engineering: Big Data & Artificial Intelligence
EFREI, Graduate School of Engineering in Computer Science & Digital Technologies
2017 - 2022 · Machine Learning specialisation

CERTIFICATIONS

- Microsoft Azure Fundamentals (AZ-900)
- HubSpot Sales Software

LANGUAGES

English: advanced / bilingual · French, Italian: native language · Spanish: bilingual